

PARKTOWN BOYS' HIGH SCHOOL



Department of Mathematics

Preliminary Examination Paper 1

22 September 2023

Subject	Mathematics
Grade	12
Marks	150
Time	3 hours

Examiner

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Moderator

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INSTRUCTIONS AND INFORMATION

1. This paper consists of 10 pages, an answer sheet and a formula sheet.
 - You may detach the formula sheet for your own use.
 - Detach the answer sheet, write your name on it and hand it in with your answers.
2. There are 12 questions.
3. Number your answers exactly like the questions in the question paper.
4. Round all answers off to TWO decimals unless indicated otherwise in a particular question.
5. An approved scientific calculator (non-programmable and non-graphical) may be used unless stated otherwise.
6. An answer alone will not necessarily be awarded full marks.
7. Look at the mark allocation as an indication of the length of the solution.
8. Write neatly and legibly.
9. Rule off after each question.

QUESTION 1

1.1 Solve for x :

1.1.1 $x^2 + 2x - 3 = 12$ (3)

1.1.2 $4x^2 + 3 = 9x$ correct to two decimal places (4)

1.1.3 $2 - x - x^2 > 0$ (4)

1.1.4 $x + 1 = \sqrt{x + 13}$ (5)

1.2 Solve for x and y simultaneously:

$$\begin{aligned}x + y &= 1 \\x^2 - y^2 - xy - y &= 13\end{aligned}\quad (6)$$

1.3 If $\frac{27^a}{9^b} = 81$ determine the value of $3a - 2b$. (3)

[25]

QUESTION 2

2.1 How many terms in the arithmetic series $-20 - 15 - 10 - \dots$ will add up to 225? (5)

2.2 Write the geometric series $-\frac{3}{2} + 3 - 6 + \dots - 96$ in sigma notation. (4)

2.3 Consider the geometric series: $1 + \frac{1}{2} + \frac{1}{4} + \dots$

2.3.1 Calculate the sum to infinity of the series. (2)

2.3.2 Determine the smallest possible value of k for which $S_\infty - S_k < \frac{1}{1000}$. (5)

[16]

QUESTION 3

A quadratic sequence with general term T_n has the following properties:

- $T_{31} = 1335$
- $T_n - T_{n-1} = 3n - 4$

Determine the general term T_n . [6]

QUESTION 4

Consider the function $f(x) = \frac{1-x}{x+2}$.

4.1 Show that $f(x) = \frac{3}{x+2} - 1$. (2)

4.2 Sketch the graph of f on the answer sheet provided. **REMEMBER TO DETACH THE ANSWER SHEET AND HAND IT IN WITH YOUR ANSWER BOOK.** (4)

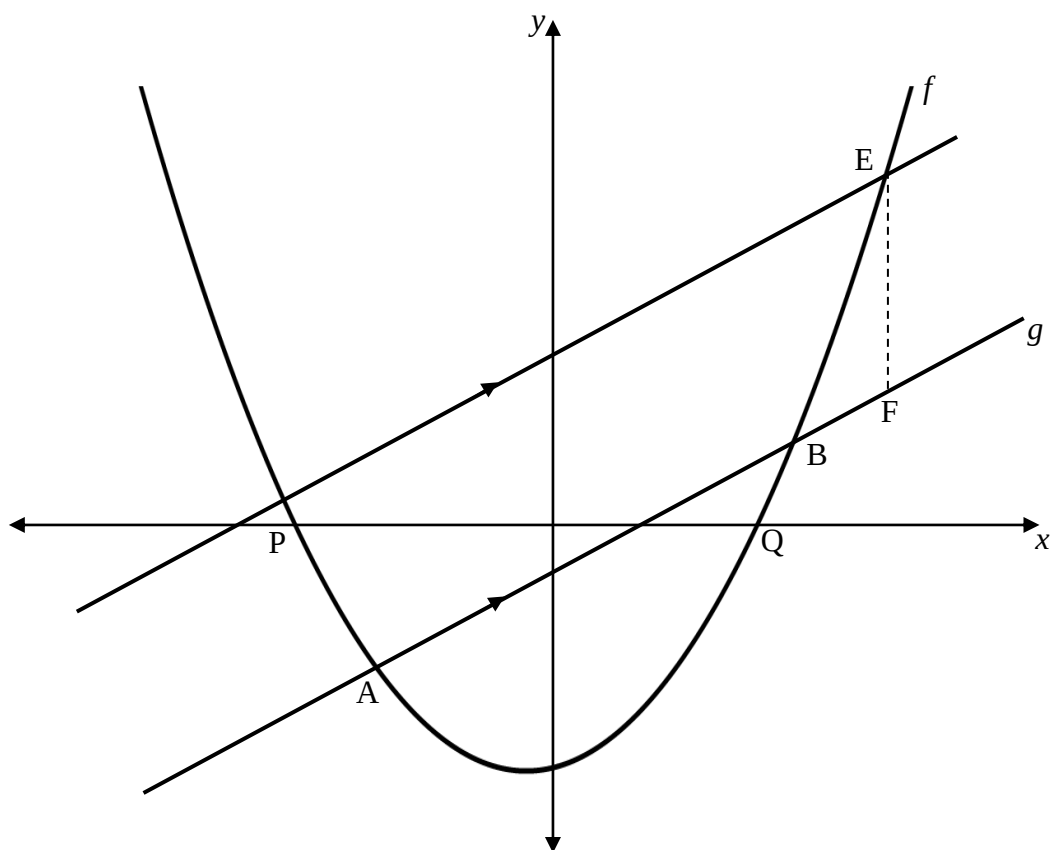
4.3 What is the range of f ? (2)

4.4 Write down the equation of the line of symmetry of f that has a negative gradient. (2)

[10]

QUESTION 5

In the diagram below, the graph of $f(x) = 2x^2 + x - 6$, $g(x) = 3x - 2$ and a line $y = mx + c$ parallel to g are sketched. P and Q are the x-intercepts of f . A and B are the points of intersection between f and g . EF is a vertical line with $EF = 8$ units.

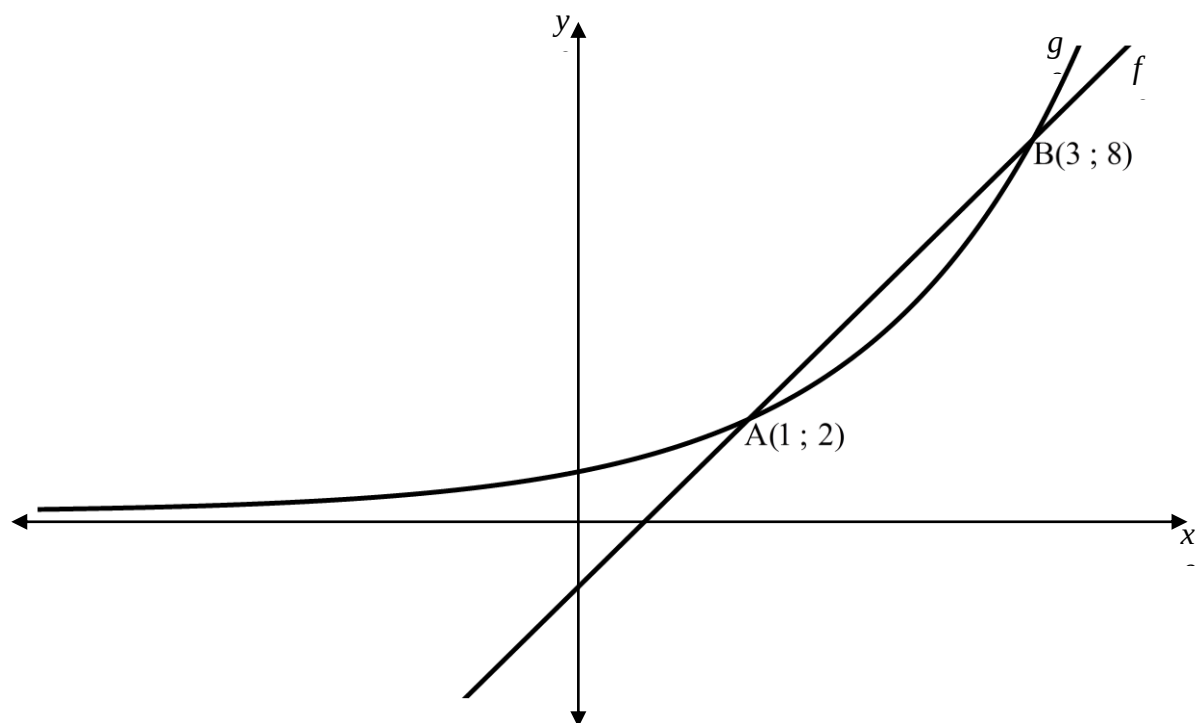


- 5.1 Determine the coordinates of the turning point of f . (3)
- 5.2 Determine the length of PQ. (4)
- 5.3 Determine the values of m and c . (2)
- 5.4 For which value(s) of x is $f(x) \geq g(x)$? (4)
- 5.5 If $f(x) = h'(x)$ determine the values of x for which h is
 - 5.5.1 increasing? (2)
 - 5.5.2 concave down? (2)
- 5.6 For which value(s) of k will the equation $2x^2 + x - 6 = 3x + k$ have only one real solution? (5)

[22]

QUESTION 6

In the diagram below, the graphs of $f(x) = 3x - 1$ and $g(x) = 2^x$ are sketched. f and g intersect at $A(1 ; 2)$ and $B(3 ; 8)$.



6.1 Determine the equation of each of the following functions:

6.1.1 $y = f^{-1}(x)$ (3)

6.1.2 $y = g^{-1}(x)$ (2)

6.2 Determine the value(s) of x for which

6.2.1 $f(x) < g(x)$. (2)

6.2.2 $g^{-1}(x) < 0$. (2)

6.2.3 $f^{-1}(x) = g^{-1}(x)$. (2)

[11]

QUESTION 7

- 7.1 Calculate the value of x if $x\%$ p.a. compounded semi-annually is equivalent to an effective rate of 15% p.a. (3)
- 7.2 James wants to save R400 000 to buy a new car in 4 years' time. He makes equal quarterly payments into a savings account, starting immediately. He makes the last payment at the end of the 4 years. How much must he save each quarter, if interest is calculated at 8% p.a. compounded quarterly? (4)
- 7.3 Kevin borrows R750 000, which he repays by means of monthly instalments of R16 000, starting one month after receiving the loan. The interest rate on the loan is 12,5% p.a. compounded monthly.
- 7.3.1 How many payments are required to repay the loan? (5)
- 7.3.2 Kevin's final payment (made one month after the last full payment) will be less than R16 000. Calculate the value of this last payment. (5)

[17]

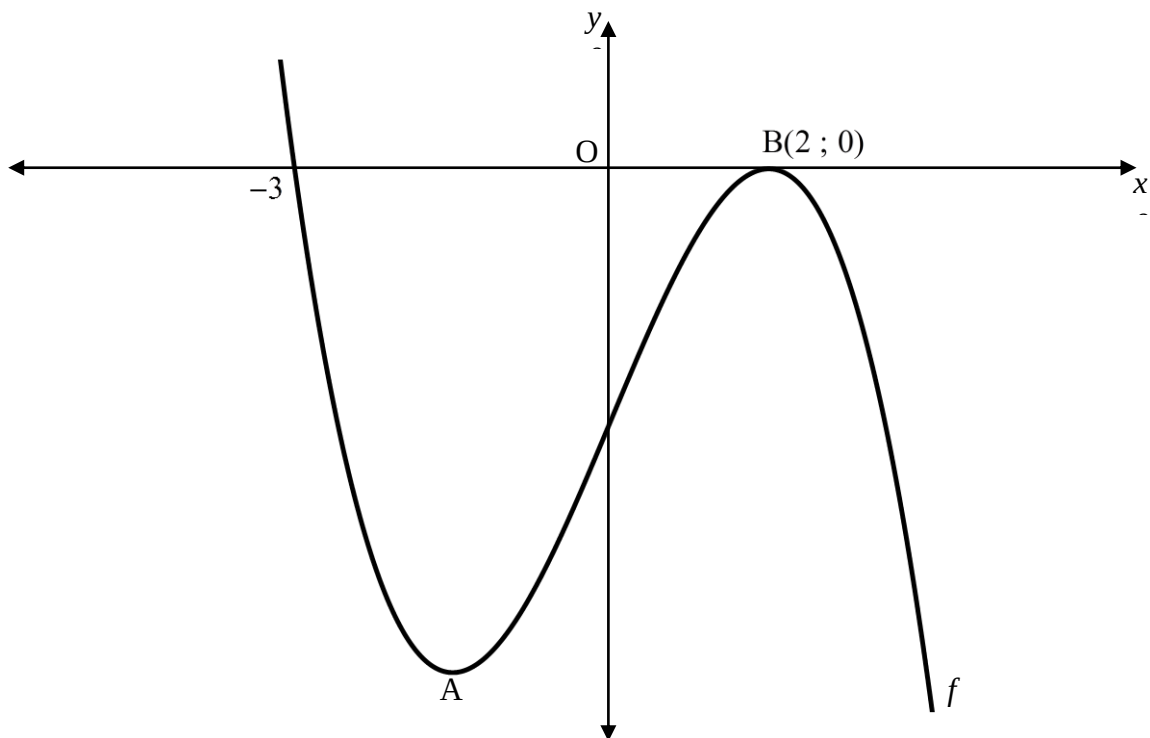
QUESTION 8

- 8.1 Determine $f'(x)$ from first principles if $f(x) = \frac{4}{x}$. (4)
- 8.2 Determine: $D_x \left[\frac{(\sqrt{x}-1)^2}{\sqrt{x}} \right]$ (4)

[8]

QUESTION 9

In the diagram below, the graph of $f(x) = -x^3 + ax^2 + bx + c$ has been sketched. A and B(2 ; 0) are turning points of f . f passes through the x -axis at $x = -3$.

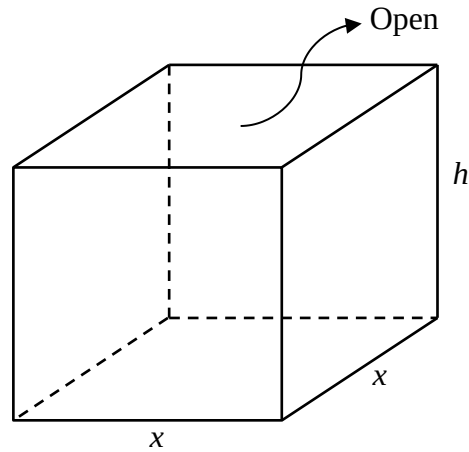


- 9.1 Show that $a = 1$, $b = 8$ and $c = -12$. (3)
- 9.2 Determine the coordinates of A. (4)
- 9.3 Determine the x -coordinate of the point of inflection of f . (3)
- 9.4 For which value(s) of x is $f(x) \cdot f'(x) < 0$? (2)

[12]

QUESTION 10

The sketch below shows a box with a square base that is open at the top. The sides of the square base are x cm and the height of the box is h cm. $1\,200\text{ cm}^2$ of material is available to create this box.



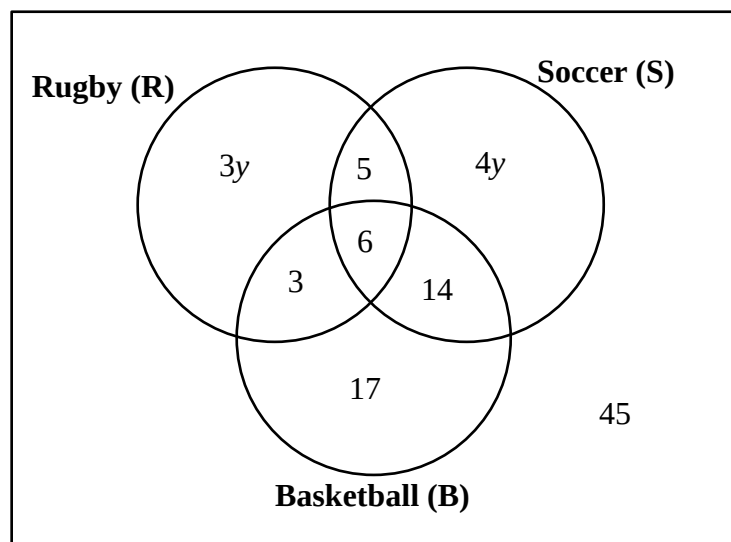
10.1 Show that the volume of the box is given by $V = 300x - \frac{x^3}{4}$. (4)

10.2 Determine the maximum volume of the box. (5)

[9]

QUESTION 11

There are 160 Grade 12's at Parktown Boys' High School. A survey was conducted to see how many boys play Rugby (**R**), Basketball (**B**) and Soccer (**S**). The results of the survey is represented in the Venn Diagram below:



11.1 Show that $y = 10$. (2)

11.2 Calculate the probability that a randomly selected boy plays

11.2.1 Rugby. (2)

11.2.2 none of these three sports. (2)

11.2.3 at least two of these three sports. (3)

[9]

QUESTION 12

The letters of the word **CENTENARY** are to be arranged. (Repeated letters are regarded as identical)

12.1 In how many possible ways can these letters be arranged? (2)

12.2 What is the probability that an arrangement will start with the letter **T** and end with the letter **E**? (3)

[5]

NAME: _____

FORM CLASS: _____

QUESTION 4.2

